DQC1-completeness of normalized trace estimation for functions of log-local Hamiltonians

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Joint work with

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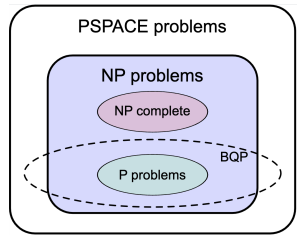
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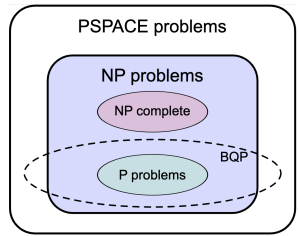
Background

- When exploring quantum capability, we are usually interested in BQP-complete problems, which capture the full power of efficient quantum computation.
- Many important quantum algorithmic tasks involve matrix functions $f(A)$:
  - Hamiltonian simulation e^{-iAt}
  - Linear systems A^{-1}
  - Gibbs states $e^{-\beta A}$



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- This talk: a restricted model (DQC1) and a natural spectral quantity, normalized trace $\frac{1}{N}\mathrm{Tr}[f(A)]$.

Problem statement

Problem (Normalized trace estimation)

Let $f : [-1, 1] \to [-1, 1]$ be continuous, A be a Hermitian matrix of dimension N with $\|A\| \leq 1$. What is the complexity of estimating $\frac{1}{N} \operatorname{Tr}[f(A)]$?

Motivation: many problems reduce to such spectral sums, e.g.,

- Machine learning: $\log \det(A) = \operatorname{Tr}[\log A]$
- Statistical physics: partition function $\operatorname{Tr}[e^{-\beta A}]$
- Graph theory: $\operatorname{Tr}[A^d]$ counts closed walks

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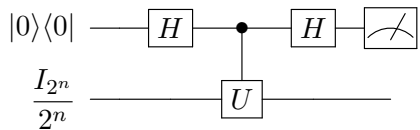
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This problem is relevant to DQC1, a restricted quantum computational model.

Question: when is this quantumly easy but classically hard?

The one-clean-qubit model (DQC1)

- It is a simple model, motivated by nuclear magnetic resonance quantum computing: one pure qubit, n maximally mixed qubits [Knill-Laflamme '98]



$$\Pr[0] = \frac{1}{2} + \frac{1}{2} \cdot \frac{\mathrm{Re}(\mathrm{Tr}[U])}{2^n}$$

- $O(1/\varepsilon^2)$ repetitions estimate the real part to additive error ε.
- Estimating $\frac{1}{2^n}\mathrm{Tr}[U]$ is DQC1-complete[1] [Knill-Laflamme '98]
- A natural framework for studying intermediate quantum computational power.

[1] DQC1-complete: problems capturing the full power of DQC1.

The one-clean-qubit model (DQC1)

- Believed weaker than BQP, but can still solve some problems for which no efficient classical algorithms are known.

Shepherd '06, Shor-Jordan '08	DQC1 = DQCk for $k = O(\log n)$
Shor-Jordan '08	Estimating Jones polynomial is DQC1-complete
Brandão '08	Estimating $\frac{1}{2^n}\mathrm{Tr}[e^{-\beta A}]$ is DQC1-complete
Morimae-Fujii-Fitzsimons '14	No efficient classical simulation unless PH collapses
Edenhofer-Hasegawa-Le Gall '25	Estimating $\frac{1}{2^n}\mathrm{Tr}[A^k]$, $\frac{1}{2^n}\mathrm{Tr}[A^{-1}]$ is DQC1-complete
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- Above results assume A is log-local: $A = \sum_k A_k$, poly(n) terms, each A_k acts nontrivially on $O(\log n)$ qubits.

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- Above results assume A is log-local: $A = \sum_k A_k$, poly(n) terms, each A_k acts nontrivially on $O(\log n)$ qubits.
- Existing completeness results: proofs tailored to specific functions
- Can we find a unified approach to these trace estimation problems?

A classical algorithm for estimating $\frac{1}{N}\operatorname{Tr}[f(A)]$

Assume Hermitian A is s-sparse and $f(x)$ is a continuous function.

- Approximate $f(x)$ by a polynomial $f(x) \approx g(x) = \sum_{k=0}^{d} a_k x^k$.
- Then
$$\frac{1}{N}\operatorname{Tr}[f(A)] \approx \frac{1}{N}\operatorname{Tr}[g(A)] = \mathbb{E}_i[(g(A))_{ii}].$$
- Sample $i \in [N]$ uniformly and compute $(g(A))_{ii} = \sum_{k=0}^{d} a_k (A^k)_{ii}$
  - $O(1/\varepsilon^2)$ samples
  - each sample costs $O(s^d)$
- Total cost: $O(s^d/\varepsilon^2)$ [Edenhofer-Hasegawa-Le Gall '25]. Efficient if $d = O(\log n)$
- So the complexity is essentially governed by how well $f(x)$ can be approximated by a low-degree polynomial.

Approximate degree

- In complexity theory, we are interested in the approximate degree:

$$\widetilde{\deg}_\varepsilon(f) = \min\{d \in \mathbb{N} : |f(x) - g(x)| \le \varepsilon \text{ for all } x \in [-1,1], \deg(g) \le d\}.$$

- Appears in the lower bound analysis of many quantum/classical algorithms.
- For constant ε, the normalized trace $\frac{1}{N}\operatorname{Tr}[f(A)]$ can be estimated using
  - Classically: $O\big(s^{\widetilde{\deg}_\varepsilon(f)}\big)$ queries
  - Quantumly: $\widetilde{O}\big(s \cdot \widetilde{\deg}_\varepsilon(f)\big)$ queries [Montanaro-Shao '24]
- Two upper bounds suggest an exponential gap in terms of $\widetilde{\deg}_\varepsilon(f)$?

Our results

Assume $f : [-1, 1] \to [-1, 1]$ is Lipschitz continuous[2], A is Hermitian with $\|A\| \leq 1$, and constant accuracy ε.

Result 1 (Query complexity): Assume A is s-sparse

	Quantum	Classical
Upper bound	$s \cdot \widetilde{\deg}_\varepsilon(f)$	$s^{\widetilde{\deg}_\varepsilon(f)}$
Lower bound	$\widetilde{\deg}_\varepsilon(f)$	$(s/2)^{\widetilde{\deg}_\varepsilon(f)/16}$

$\Rightarrow$ **This confirms the exponential separation.**

Result 2 (DQC1-completeness): Assume A is log-local and $\widetilde{\deg}_\varepsilon(f) = \Omega(\text{polylog}\, N)$, then estimating $\frac{1}{N}\operatorname{Tr}[f(A)]$ is DQC1-complete.

$\Rightarrow$ **This unifies all previous completeness results for specific functions.**

[2] We prove this under a technical approximation condition. Lipschitz version: follow-up note [Shao '26]

Technical overview

Main tools:

- Periodic Jacobi matrix
- Chebyshev equioscillation theorem
- Feynman's circuit-to-Hamiltonian construction
- Trace version (cyclic analogue) of the k-Forrelation problem

Technical overview

n-qubit unitary $U = U_m \cdots U_2 U_1$, $m = \mathrm{poly}(n)$, each U_t acts on $O(\log n)$ qubits.

- Feynman's circuit-to-Hamiltonian construction: ($|m\rangle := |0\rangle$)

$$A = \sum_{t=1}^{m} a_t |t-1\rangle\langle t-1| \otimes I_n + \sum_{t=1}^{m} b_t \left(|t\rangle\langle t-1| \otimes U_t + |t-1\rangle\langle t| \otimes U_t^\dagger \right),$$

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- A key observation: for any eigenpair $(e^{i\theta}, |\phi_\theta\rangle)$ of U, the restriction of A to the invariant subspace spanned by

$$\Big\{ |0\rangle \otimes |\phi_\theta\rangle, \quad |1\rangle \otimes U_1 |\phi_\theta\rangle, \quad \ldots, \quad |m-1\rangle \otimes U_{m-1} \cdots U_2 U_1 |\phi_\theta\rangle \Big\}$$

is a periodic Jacobi matrix of the form

$$A_\theta := \begin{bmatrix} a_1 & b_1 & & & b_m e^{i\theta} \\ b_1 & a_2 & b_2 & & \\ & b_2 & \ddots & \ddots & \\ & & \ddots & \ddots & b_{m-1} \\ b_m e^{-i\theta} & & & b_{m-1} & a_m \end{bmatrix}.$$

Technical overview

- Thus

$$\mathrm{Tr}[f(A)] = \sum_{\theta:\, e^{i\theta} \text{ is an eigenvalue of } U} \mathrm{Tr}[f(A_\theta)].$$

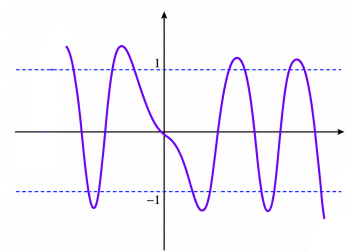
- Recall estimating $\frac{1}{N}\mathrm{Re}(\mathrm{Tr}[U]) = \frac{1}{N}\sum_\theta \cos\theta$ is DQC1-complete.

Technical overview

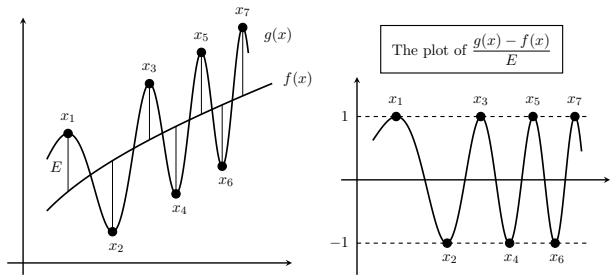
- Thus
$$\mathrm{Tr}[f(A)] = \sum_{\theta:\, e^{i\theta} \text{ is an eigenvalue of } U} \mathrm{Tr}[f(A_\theta)].$$
- Recall estimating $\frac{1}{N}\mathrm{Re}(\mathrm{Tr}[U]) = \frac{1}{N}\sum_\theta \cos\theta$ is DQC1-complete.
- About periodic Jacobi matrix,
$$\det(xI - A_\theta) = 2b_1 \cdots b_m\big(\Delta(x) - \cos\theta\big),$$
- $\Delta(x)$ is independent of θ and $\mathrm{Tr}[\Delta(A_\theta)] = m\cos\theta$.

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$$\mathrm{Tr}[f(A)] = \sum_{\theta:\, e^{i\theta} \text{ is an eigenvalue of } U} \mathrm{Tr}[f(A_\theta)].$$
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- About periodic Jacobi matrix,
$$\det(xI - A_\theta) = 2b_1 \cdots b_m\big(\Delta(x) - \cos\theta\big),$$
- $\Delta(x)$ is independent of θ and $\mathrm{Tr}[\Delta(A_\theta)] = m\cos\theta$.
- A natural but "incorrect" idea: if $\Delta(x) \approx f(x)$, then $\mathrm{Tr}[f(A_\theta)] \approx m\cos\theta$.
(e.g. Chebyshev polynomial $f(x) = T_m(x)$)
- But $\Delta(x)$ is an oscillatory polynomial.



Technical overview



- (crucial) Using Chebyshev equioscillation theorem to fix it.
- Roughly, $\Delta(x) \approx \dfrac{f(x)-g(x)}{E}$, from which we can reconstruct A.
- Then we obtain a connection between $\frac{1}{mN}\operatorname{Tr}[f(A)]$ and $\frac{1}{N}\operatorname{Re}(\operatorname{Tr}[U])$.
- $\Rightarrow$ Result 2: DQC1-completeness.

Classical query lower bound: k-cyclic Forrelation problem

- Given $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_k \in \{-1,1\}^N$ with oracles $O_{\boldsymbol{x}_1}, \ldots, O_{\boldsymbol{x}_k}$, and $H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$.
- Let $U = O_{\boldsymbol{x}_1} H^{\otimes n} O_{\boldsymbol{x}_2} H^{\otimes n} \cdots H^{\otimes n} O_{\boldsymbol{x}_k} H^{\otimes n}$ and

$$\texttt{Trace}_k(\boldsymbol{x}) := \frac{1}{2^n} \operatorname{Tr}[U].$$

- Recent results: lower bounds of classical query complexity of estimating $\texttt{Trace}_k(\boldsymbol{x})$ to constant additive error [Zhang-Shao '26, arXiv:2610.41694]
  - Nonadaptive $\Omega_k(N^{1-1/\lfloor k/2 \rfloor})$.
  - Adaptive $\Omega(N^{1/4})$ for $k \geq 4$.
- Using the connection established above, we can prove a lower bound of

$$\widetilde{\Omega}((s/2)^{\widetilde{\deg}_\varepsilon(f)/16}).$$

- $\Rightarrow$ Result 1: Classical query lower bound.

Conclusion

Problem: Estimating the normalized trace $\frac{1}{N}\operatorname{Tr}[f(A)]$ for log-local Hamiltonians.

- We establish a unified framework for proving DQC1-completeness.
- Approximate degree is a key parameter governing the complexity:
  - Efficient DQC1 algorithms
  - Classical query lower bound

Thanks very much for your attention!

Technical condition on the polynomial approximation

- Our results hold when $E_d/E_{d-1} < 1/2$, where $d = \widetilde{\deg}_\varepsilon(f)$ and

$$E_d := \min_{\deg(g)\le d} \max_{-1\le x\le 1} |f(x) - g(x)|$$

- Many functions satisfy this condition.
- This condition can be relaxed to Lipschitz continuity. [Shao '26]